Recap: Topic 5

Up to this point in Topic 5, we

- Have learnt about how initial boundary value problems are used to model real-world phenomena, particularly those involving the heat equation
- Applied separation of variables to the 1D and 2D heat equations (+ Laplace eqn)
- For homogeneous BCs, found that the solution forced by an inhomogeneous IC is a superposition (with time-dependent coefficients) of eigenfunctions of the spatial problem (non-trivial solutions of homogeneous problem)
- For inhomogeneous BCs, the solution involves steady-state and transient components

Lecture 25

In Lecture 25, we will:

1. Assess whether one of our SoV solutions is genuine
2. Discuss weak solutions (not assessed)
3. Discuss uniqueness of solutions (not assessed)

On the informality of formal solutions

Recap: Formal and genuine solutions of ODEs (Topic 2)

1. Formal solution: term by term differentiated series satisfies the ODE
2. Genuine solution: series and all derivatives up to order of ODE converge uniformly

We have similar concepts for PDEs.

- We used SoV to find *formal solutions*. We were not concerned with whether the series solution converged and over what domain.
- To show a solution to a PDE is a *genuine solution*, i.e., that it satisfies the PDE, we have to show we can bring all derivatives inside the infinite sum.

Recap: Uniform convergence of a series

Let $w_n(t)$ $(n = 1, 2, 3, \ldots)$ be functions on D.

The series $\sum_{n=1}^{\infty} w_n(t)$ converges uniformly to $W(t)$ on D if

$$\lim_{N\to\infty} \max_{t\in D} \left| \sum_{n=1}^{N} w_n(t) - W(t) \right| = 0$$

Recap: Weierstrass test for uniform convergence

- Let $\max_{t\in D} |w_n(t)| \leqslant a_n$ for some $a_n \in \mathbb{R}$ such that $\sum_{n=1}^{\infty} a_n$ converges
- Then $\sum_{n=1}^{\infty} w_n(t)$ converges *uniformly* on D

Recap: Differentiating a series

- Let $P(t) = \sum_{n=0}^{\infty} w_n(t)$ where w_n are continuous differentiable functions on D
- Suppose $P(t)$ converges uniformly on D
- Suppose $\sum_{n=0}^{\infty} w_n'(t)$ converges uniformly on D
- Then $P(t)$ can be differentiated term by term, i.e.,

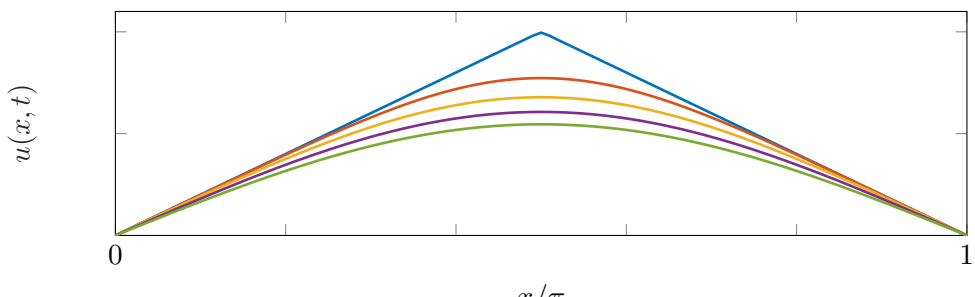
$$\frac{\mathrm{d}P}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \sum_{n=0}^{\infty} w_n(t) = \sum_{n=0}^{\infty} \frac{\mathrm{d}}{\mathrm{d}t} w_n(t)$$

Exercise 5.3: 1D heat equation Prb 1 revisited

We found the temperature along a bar with Dirichlet BC and a prescribed IC as

$$u(x,t) = -\frac{4L}{\pi^2}\sum_{m=1}^{\infty}\underbrace{\frac{(-1)^m}{(2m-1)^2}\mathrm{e}^{-\alpha k_{2m-1}^2 t}\sin(k_{2m-1}x)}_{u_m(x,t)} \quad \text{where} \quad k_n = \frac{n\pi}{L}$$

$\alpha = 1\,\mathrm{mm}^2\,\mathrm{s}^{-1}$ (quartz) and $L = \pi\,\mathrm{mm}$



Consider the domain

$$D = \{(x,t) \in \mathbb{R}^2 : x \in [0,L] \quad \text{and} \quad t \geqslant T\}$$

for some $T > 0$.

Consider

$$u_m(x,t) = \frac{(-1)^m}{(2m-1)^2} \mathrm{e}^{-\alpha k_{2m-1}^2 t} \sin(k_{2m-1}x)$$

We can bound u_m by

$$\max_{(x,t)\in D} |u_m(x,t)| = \frac{1}{(2m-1)^2} \mathrm{e}^{-\frac{\alpha\pi^2(2m-1)^2 T}{L^2}} \leqslant \frac{1}{(2m-1)^2} = a_m$$

Now

$$\sum_{m=1}^{\infty} a_m = \frac{1}{1^2} + \frac{1}{3^2} + \dots$$

converges, as it is a subset of the harmonic $p = 2$ series

Weierstrass test tells us that

$$\sum_{m=1}^{\infty} u_m(x,t)$$

converges uniformly.

We can differentiate u_m with respect to x to find

$$\frac{\partial}{\partial x} u_m(x,t) = \frac{(-1)^m k_{2m-1}}{(2m-1)^2} \mathrm{e}^{-\alpha k_{2m-1}^2 t} \cos(k_{2m-1} x)$$

We simplify the expression to

$$\frac{\partial}{\partial x} u_m(x,t) = \frac{(-1)^m \pi}{L\,(2m-1)} \mathrm{e}^{-\alpha k_{2m-1}^2 t} \cos(k_{2m-1} x)$$

We can bound $\frac{\partial}{\partial x} u_m$ by

$$\max_{(x,t)\in D} \left| \frac{\partial}{\partial x} u_m(x,t) \right| = \frac{\pi}{L\,(2m-1)} \mathrm{e}^{-\frac{\alpha\pi^2(2m-1)^2 T}{L^2}} \leqslant \frac{\pi}{L} \mathrm{e}^{-\frac{\alpha\pi^2(2m-1)^2 T}{L^2}} = a_m$$

Now

$$\sum_{m=1}^{\infty} \frac{\pi}{L} \mathrm{e}^{-\frac{\alpha \pi^2 (2m-1)^2 T}{L^2}}$$

converges, e.g., by ratio test.

It follows that

$$\sum_{m=1}^{\infty} \frac{\partial}{\partial x} u_m(x,t) \quad \text{converges uniformly}$$

and hence term-by-term differentiation wrt x of

$$\sum_{m=1}^{\infty} u_m(x,t) \quad \text{is justified}$$

We note that our estimate required $t \geqslant T > 0$. Indeed, uniform convergence fails on the larger domain $t > 0$.

We can differentiate u_m twice with respect to x to find

$$\frac{\partial^2}{\partial x^2} u_m(x,t) = -\frac{(-1)^m k_{2m-1}^2}{(2m-1)^2} \mathrm{e}^{-\alpha k_{2m-1}^2 t} \sin(k_{2m-1} x)$$

We simplify the expression to

$$\frac{\partial^2}{\partial x^2} u_m(x,t) = \frac{(-1)^m \pi^2}{L^2} \mathrm{e}^{-\alpha k_{2m-1}^2 t} \sin(k_{2m-1} x)$$

We can bound $\frac{\partial^2}{\partial x^2} u_m$ by

$$\max_{(x,t)\in D} \left| \frac{\partial^2}{\partial x^2} u_m(x,t) \right| = \frac{\pi^2}{L^2} \mathrm{e}^{-\frac{\alpha \pi^2 (2m-1)^2 T}{L^2}} = a_m$$

Now

$$\sum_{m=1}^{\infty} \frac{1}{L^2} \mathrm{e}^{-\frac{\alpha \pi^2 (2m-1)^2 T}{L^2}}$$

converges, e.g., by ratio test.

It follows that

$$\sum_{m=1}^{\infty} \frac{\partial^2}{\partial x^2} u_m(x,t) \quad \text{converges uniformly}$$

and hence term-by-term differentiation wrt x of

$$\sum_{m=1}^{\infty} \frac{\partial}{\partial x} u_m(x,t) \quad \text{is justified}$$

We can differentiate u_m with respect to t to find

$$\frac{\partial}{\partial t}u_m(x,t) = -\alpha\frac{(-1)^m k_{2m-1}^2}{(2m-1)^2}\mathrm{e}^{-\alpha k_{2m-1}^2 t}\sin(k_{2m-1}x)$$

Note that

$$\frac{\partial}{\partial t}u_m = \alpha\frac{\partial^2}{\partial x^2}u_m$$

and hence we can repeat the argument for uniform convergence.

Thus, term-by-term differentiation wrt t of

$$\sum_{m=1}^{\infty} u_m(x,t) \quad \text{is justified}$$

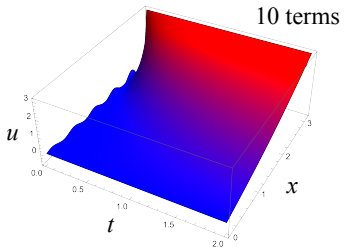
We conclude that $u(x,t)$ is a genuine solution of the PDE for any $t > 0$

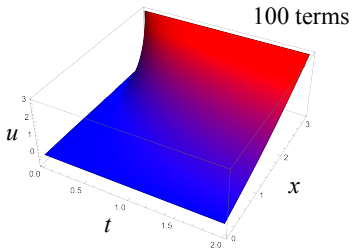
Recall: 1D heat equation, Problem 3

We noted Gibbs phenomenon at $t = 0$ for the formal solution

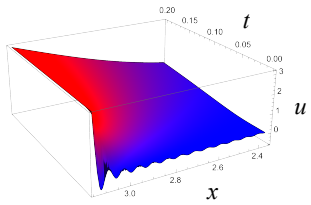
$$u(x,t) = u_S(x) + u_T(x,t) = x + \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\mathrm{e}^{-\alpha\, k_n^2\, t}\sin\left(k_n\, x\right)$$

$$\alpha = 1, \quad L = \pi$$





Zoom in on initial condition 100 terms



We can differentiate

$$u_n(x,t) = \frac{(-1)^n}{n} \mathrm{e}^{-\alpha\, k_n^2\, t} \sin\left(k_n\, x\right)$$

with respect to x and t.

As before, the exponential factor will ensure uniform convergence as long as $t \geqslant T > 0$.

This example shows pictorially why it is important to use $T > 0$ to stay away from $t = 0$.

Weak solutions (not assessed)

- Heat equation (parabolic PDE) smooths out troublesome corners and discontinuities in IC, whereas wave equation (hyperbolic PDE) propagates them
- Don't just discard formal solutions that don't have uniform convergence properties
- Formal solutions which don't satisfy the IBVP are an example of *weak solutions*
- Weak solutions are extremely useful in both numerical and analytical analysis of differential equations. You will meet them again in advanced differential equations courses.

Uniqueness (not assessed)

- After proving a solution satisfies the PDE (genuine solution) we also need to consider if it is unique, i.e., the only solution to the IBVP.

- All PDEs covered in MAST20030 produce unique solutions where the initial and boundary conditions allow a genuine solution.

- Idea of proof is to show that given two solutions, u_1 and u_2, then $v = u_1 - u_2$ must be zero everywhere. This is often done through an *energy functional*

$$E[v](t) = \int_D v(x,t)^2 \mathrm{d}x$$

where D is the spatial domain. If $E[v](t) = 0$ for all t then the solution is unique.